High-Accuracy Laser Altimetry Global Elevation Control Point Dataset for Satellite Topographic Mapping

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***Abstract*— As the accuracy of satellite laser altimetry is susceptible to real-time atmospheric conditions, within footprint topography fluctuation, detector noise, and so on, we proposed a method by comprehensively analyzing the laser ranging error and evaluation labels to extract high-accuracy elevation control points that is suitable for satellite imagery-based topographic mapping applications. Using the ICESat laser altimetry data, a global high-accuracy laser altimetry dataset, including more than 60 million control points, based on the laser altimetry ranging error model and waveform quality analysis is proposed in this article. For land areas, except for areas of water, snow/ice, and polar ice sheets, the dataset can provide the elevation control points for worldwide satellite topographic mapping using high spatial resolution imageries or other science research that depend on accurate Earth's elevation information. We further used airborne LiDAR data from six study areas around the world to carefully validate the datasets' accuracy. The results showed that this dataset can meet the accuracy requirement of global mapping using high spatial resolution satellite imageries in terrains with a slope below 25°. Compared to the raw dataset, the proportion of footprint elevations that conform to the accuracy standard (0.5 m @ slope < 2°, 1.5 m @ 2° ≤ slope < 6°, and 3 m @ 6° ≤ slope < 25°) is increased from 68.24%, 59.97%, and 26.52% to 87.58%, 90.04%, and 83.91%, respectively. This method can assure that its extracted results' accuracy is either very close to or better than that obtained by the methods proposed in relevant studies, with a much larger number of laser footprints (LFs) that have been reserved.**

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***Index Terms*— Elevation control point, evaluation label, ICESat, laser altimetry, ranging error model, topographic mapping.**

I. Introduction

HIGH-ACCURACY elevation information is important for many scientific applications, such as polar glacier monitoring, terrain deformation monitoring, and topographic mapping. Airborne LiDAR can observe the Earth's surface with high accuracy and density, but the high cost of data acquisition and the subsequent processing mean that this approach has difficulty in global-scale applications [1]. With spaceborne laser altimetric observations, the data can be expected to provide high-accuracy elevation information worldwide.

Satellite stereo imaging technology can also provide topographic maps. Multiple mapping standards have proposed the corresponding accuracy requirements for the topographic map using imageries with different ground sampling distances (GSDs) [2]–[5]. Taking the topographic map using the meter-level GSD as an example (1.25–10 m), the horizontal accuracy (RMSE) is required to be 3–14 m, and the elevation accuracy (RMSE) is required to be 0.5–6 m (at different slope areas). However, due to the influence of the base-to-height ratio, platform stability, and so on, this technology is difficult to achieve the above accuracy without ground control points, especially the elevation accuracy [6], [7]. The ground control points are mainly collected manually, which has low efficiency and is difficult to carry out in a large area of the world. Spaceborne laser altimetric observations can provide high-precision elevation information worldwide, which is beneficial to satellite stereo imaging technology to generate high-quality topographic maps worldwide.

As early as the 20th century, spaceborne laser altimetry technology was utilized for deep space exploration for mapping the terrain or supporting image stereo mapping. Subsequently, the technology was extended to be applied in the Earth observation [8]–[10]. The laser altimeter carried on the ICESat mission with the 2003–2010 operation was the first nonexperimental laser ranging instrument for observing the Earth continuously [8], [11]. Its main scientific goal is polar glacier monitoring [8], but it has also achieved a wide range of applications in other scientific fields, including topographic mapping and biomass estimation. The GLAS instrument could

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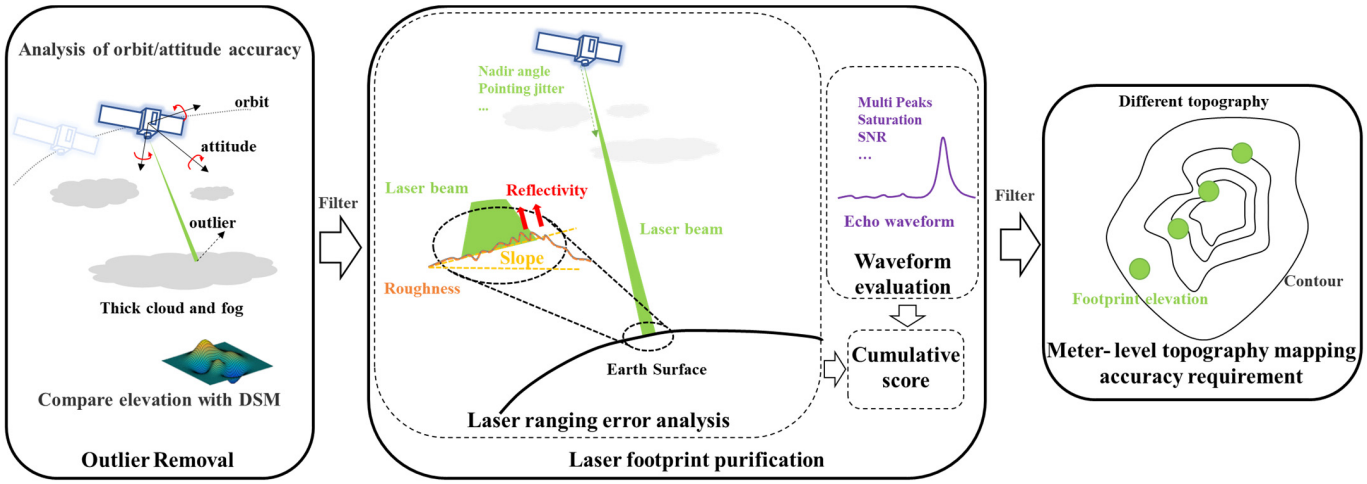


Fig. 1. Schematic of LF elevation control points extraction based on the ranging error model and evaluation labels.

emit one laser pulse (1064 nm for ranging) at a laser emit frequency of 40 Hz to detect the ground, forming a ground spot of ∼65 m in size and a ∼172-m along-ground track interval [8].

However, in the ranging and range-elevation processing, there are some influencing factors, such as atmospheric conditions, attitude accuracy, instrument noise, and within footprint roughness, resulting in an uncertain laser footprint (LF) elevation accuracy [11]. The previous studies extracting high-accuracy LFs can be used as elevation control points for topographic mapping, which mainly includes the use of the product quality labels [11], [12], reference DEM or airborne LiDAR [13]–[15], learning-based classification methods [16], or their combinations [17]–[19]. Some of the above methods have also been used to construct the global elevation control point dataset [19]. The main source of these datasets is the ICESat altimetry data. However, the majority of these methods can only eliminate the impacts of the surface reflectance and atmospheric effect [18]. They are not suitable for extension to regions with higher slopes or artificial objects, or vegetations. Moreover, the overall filtering criterion applied in these kinds of methods is too strict that it leads to fewer available data, which means that the spatial distribution in some scenes is sparse, resulting in low mapping accuracy [18], [19]. The learning-based method is reliant on the spatial distribution of the training samples, and it is difficult to apply globally [16], [20]. Therefore, a new method is required to extract accurate LF, to adapt to the satellite imagery-based topographic mapping applications in different terrains, especially for the regions where ground control points are difficult to acquire.

To overcome such limitations, there are two contributions in this article. The first, a method of extracting high-accuracy elevation control points by comprehensively analyzing the laser ranging error and evaluation labels, is proposed. The second, a global high-accuracy laser altimetry dataset using ICESat as experimental data, is proposed. To verify the accuracy of the dataset, we used airborne LiDAR data from six regions of the world. The verified results indicate that the extracted results by the proposed method can generally conform to the accuracy standards with a much better performance of footprint retention rate than previous studies. The vertical accuracy (RMSE) of this dataset can reach 0.5–3 m (different slope areas), which satisfies the accuracy requirements of stereo mapping using meter-level GSD remote sensing imageries [2]–[5]. The dataset can provide more than 60 million elevation control points within the global land area to users for large-scale topography mapping or other scientific research.

The materials and methodology adopted in this study are given in Sections II and III, respectively. The results and discussions are described in Section IV, including the accuracy verification of the extracted high-accuracy LF elevation results and a description of the high-accuracy global elevation control point dataset that we built in this study. The conclusions of this article are provided in Section V.

II. Methodology

As shown in Fig. 1, the main factors affecting the laser altimetry accuracy include the atmospheric conditions, orbit/attitude miscalculation, saturated echoes, instrument noise, within footprint slope, and surface roughness [11]–[15], [17], [18]. According to these factors, the main process is divided into two parts.

First, remove outliers due to atmospheric scattering or other situations. These outliers might have tens or even hundreds of meters difference from the ground surface, which can be easily eliminated [18]. The orbit/attitude data were also analyzed to remove low-quality LFs. Reference DSM is used to eliminate the LF outliers with elevation difference larger than 50 m [14], [15], [18]. LFs with attitude accuracy labeled as 100 in the GLAH14 dataset were also removed as a small error of the orbit/attitude may induce large footprint elevation changes in steep areas [18], [32].

Second, the laser ranging error was estimated through a ranging error model and combined with several waveform quality labels to purify the LFs. As the ranging error is only a part of the elevation error of LF. Other factors, such as pointing/attitude error, satellite orbit error, atmospheric delay correction, and tide correction, also contribute to the final LF

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accuracy. These errors are corrected and filtered by the corresponding outlier labels in the GLAS data production, and their impacts on elevation errors can be ignored. Therefore, it can use elevation accuracy requirements in different topography areas as the judging criterion for ranging error.

A. Laser Footprint (LF) Purification

1) Theoretical Laser Ranging Error Model: The laser ranging error of LF is evaluated by a simplified laser ranging error model. The simplified laser ranging error model [33] is shown in the following formula:

$$d\rho = \frac{c}{2} * \sqrt{d\Gamma} \tag{1}$$

$$\begin{aligned} d\Gamma = {} & \frac{F\left(\sigma_l^2+\sigma_h^2\right)}{\langle N\rangle} + \left(\frac{F}{\langle N\rangle}+\frac{1}{K_s}\right)\cdot\frac{4\left(\mathrm{std}(\Delta\xi)^2\right)\cos^2 S}{c^2\cos^2(\phi+S)} \\ & + \left(\frac{F}{\langle N\rangle}+\frac{1}{2K_s}\right)\frac{4z^2\tan^2\theta_T}{c^2\cos^2\phi} \\ & \cdot\left[\tan^2\theta_T+\tan^2(\phi+S)+\frac{\tan^2 S\cos^2 S}{\cos^2(\phi+S)}\right] \\ & + \frac{4z^2\left(1+\tan^2\theta_T\right)^2}{c^2\cos^2\phi} \\ & \cdot\left[\tan^2(\phi+S)(\mathrm{std}(\Delta\phi)^2)+\frac{\tan^2 S\cos^2 S\cos^2\phi}{\cos^2(\phi+S)}\cdot(\mathrm{std}(\Delta\phi)^2)\right] \end{aligned} \tag{2}$$

where $d\rho$ is the satellite laser ranging error; σ_t is the pulsewidth's rms; σ_h is the impulse-response width's rms (receiver), which can be calculated according to the following equation; $\langle N\rangle$ is the total detected photon count, which can be estimated according to the following equation; K_S indicates the area percentage of the receiving surface relative to the speckled surface, which can be calculated according to the following equation; c, F, and z are the speed of light, detector's excess noise factor, and orbit altitude, respectively; ϕ is the beam nadir angle; θ_T is the laser beam divergence, which can be estimated by the following equation; S is the slope that locates in the LF; $\mathrm{std}(\Delta\xi)$ is the surface roughness that locates in the LF; and $\mathrm{std}(\Delta\phi)$ is the pointing error

$$\sigma_h = \frac{\Delta t}{\sqrt{12}} \tag{3}$$

$$\langle N\rangle = \frac{E}{h\upsilon} \tag{4}$$

$$K_S = \pi A\left(\frac{2\tan\theta_T}{\lambda}\right)^2 \tag{5}$$

$$\theta_T \approx a\tan\left(\frac{a}{\rho}\right) \tag{6}$$

where Δt is the sampling period; E and $h\upsilon$ are the echo energy (EN) and the photon energy, respectively; A indicates the receiving aperture area; λ indicates the pulse wavelength; a is half of the LF's major axis; and ρ is the ranging value. According to the characteristics of the ICESat altimeter, σ_t, Δt, F, λ, and A were set to 6 ns, 1 ns, 3.2, 1064 nm, and 1 m^2, respectively [8]. The slope in the LFs is calculated by adopting the methodology proposed in [34]. The roughness and EN can be characterized by the algorithm proposed in [16].

From (1), the satellite laser ranging error, i.e., $d\rho$, can be further decomposed into the system error (e_{System}), surface roughness error (e_{SR}), beam curvature error (e_{BC}), nadir angle and surface slope error ($e_{N\&S}$), and pointing jitter error (e_{PJ}). These suberrors can be calculated as shown in the following equations:

$$e_{\mathrm{System}} = \frac{c}{2} * \sqrt{\frac{F\left(\sigma_l^2+\sigma_h^2\right)}{\langle N\rangle}} \tag{7}$$

$$e_{\mathrm{SR}} = \frac{c}{2} * \sqrt{\left(\frac{F}{\langle N\rangle}+\frac{1}{K_s}\right)\frac{4(\mathrm{std}(\Delta\xi)^2)\cos^2 S}{c^2\cos^2(\phi+S)}} \tag{8}$$

$$e_{\mathrm{BC}} = \frac{c}{2} * \sqrt{\left(\frac{F}{\langle N\rangle}+\frac{1}{2K_s}\right)\cdot\frac{4z^2\tan^2\theta_T}{c^2\cos^2\phi}\tan^2\theta_T} \tag{9}$$

$$e_{N\&S} = \frac{c}{2} * \sqrt{\begin{array}{l}\left(\frac{F}{\langle N\rangle}+\frac{1}{2K_s}\right)\frac{4z^2\tan^2\theta_T}{c^2\cos^2\phi} \\ \cdot\left(\tan^2(\phi+S)+\frac{\tan^2 S\cos^2 S}{\cos^2(\phi+S)}\right)\end{array}} \tag{10}$$

$$e_{\mathrm{PJ}} = \frac{c}{2} * \sqrt{\begin{array}{l}\frac{4z^2\left(1+\tan^2\theta_T\right)^2}{c^2\cos^2\phi} \\ \cdot\left[\tan^2(\phi+S)(\mathrm{std}(\Delta\phi)^2)\right. \\ \qquad \left.+\frac{\tan^2 S\cos^2 S\cos^2\phi}{\cos^2(\phi+S)}\cdot(\mathrm{std}(\Delta\phi)^2)\right].\end{array}} \tag{11}$$

2) Judging Criterion of Subitem of Laser Ranging Error: Among the five suberrors, the contribution of the beam curvature error is very small [33], so this error was not further analyzed.

The system error is not affected by terrain. Considering that the nominal vertical error (RMSE) of ICESat after calibration is about 14 cm, the system error (RMSE) should be no higher than 10 cm. Ranging error is a part of elevation error of LF. Therefore, it can use elevation accuracy requirements in different topography areas as judging criterion for e_{SR}, $e_{N\&S}$, and e_{PJ}. Here, each error should be no longer than the threshold, i.e., half of the elevation error of requirements.

To avoid a high missing alarm rate, which can be caused by the setting of the elevation error value, i.e., the nadir angle and surface slope error and pointing jitter error, as the two errors make major contribution among the five error items, these should be both close to but not exceeding the thresholds. Further analysis is needed for this situation. Here, when $e_{N\&S} \le$ half of elevation error or $e_{\mathrm{PJ}} \le$ half of elevation error, the sum of these two errors should be no higher than the 75% of elevation error.

3) Waveform Quality Analysis Using Evaluation Labels: As a useful supplement for the evaluation of LF quality, further detailed analysis of the laser waveform is required. According to related research, the waveform can be analyzed mainly from the number of echo peaks, echo gain, SNR, and echo saturation [8], [11], [16], [18]. The peak number represents the variability of vertical direction within the LF, and the greater the peak number, the greater the vertical variation. Echo gain identifies

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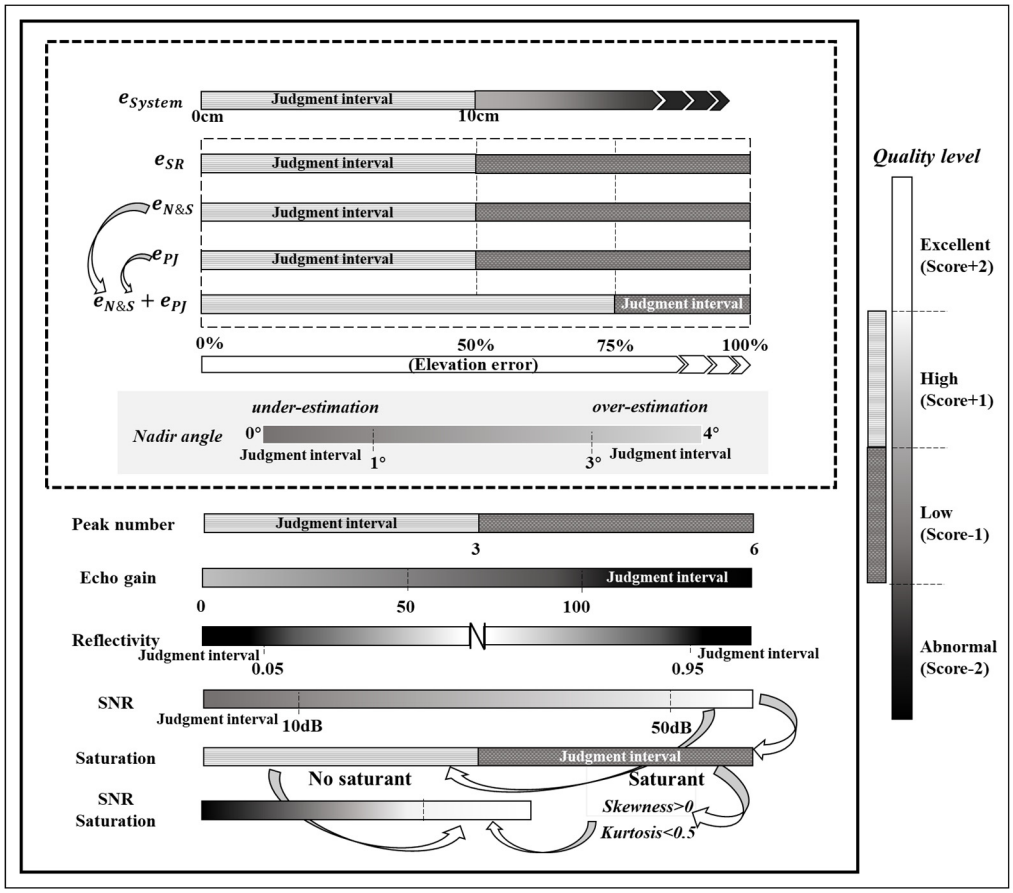


Fig. 2. Criterion of cumulative score for each suberror and waveform quality evaluation labels.

LFs affected by the energy-reducing phenomenon such as atmospheric particulate scattering. Reflectivity identifies the LFs affected by both scattering (atmospheric) and saturation, i.e., the reflectivity of the LF affected by echo saturation should be high, while the LF affected by atmospheric scattering should be low. SNR is used to detect the noise from the instruments. Echo saturation reduces the accuracy of the echo parameter calculation, which leads to a reduction in the ranging accuracy.

4) Judging Criterion of Evaluation Labels: Based on related research [8], [11], [16], [18], we made the judging criteria for the waveform quality.

As the number range of the echo peaks of ICESat is from 1 to 6, the LF with the number of peaks $\leq$3 has lower vertical variation. Since the range of echo gain is mainly focused on $\leq$50, the LF with the echo gain $\geq$100 is extremely abnormal. The LF's reflectivity value is generally in the 0.05–0.95 range, so the LF beyond this range is extremely abnormal. The LF with SNR $\leq$10 dB indicates that the noise is higher than the valid signal, and this LF is lower quality. The LF with a saturated echo is of lower quality.

Moreover, the LF with a high SNR needs further analysis. The LF with a nonsaturated echo and high SNR is excellent quality. If the LF has a saturated echo and high SNR but a good echo symmetry, this LF is also excellent quality. Here, the saturated echo with kurtosis >0 and echo skewness <0.5 is considered a good echo symmetry.

5) Cumulative Score: The ranging error of LF can be estimated by using the theoretical laser ranging error mode. If this ranging error is less than the elevation error, the LF was reserved and then cumulative scored. The criterion of cumulative score is shown in Fig. 2. Each item in the criterion complies with the judging criteria of suberror and waveform evaluation labels and then is scored according to the quality level. Moreover, the roughness, slope, echo energy, and other measured values can also have errors, and these errors can lead to overestimation or underestimation in the calculation of the suberrors, due to the size of the beam nadir angle. Therefore, this angle needs to be analyzed. As the nadir angle of ICESat is mainly between 0° and 4°, these errors with the nadir angle $\leq$1° are underestimation, and these errors with the nadir angle $\geq$3° are overestimation.

For each footprint elevation, its various items (ranging error or waveform evaluation label) are first judged whether it is located in the judgment interval. Then, the corresponding items located in the judgment interval are scored. The score can refer to the quality level in Fig. 2, including excellent ($+2$), high ($+1$), low (-1), and abnormal (-2) four different levels. Finally, the score values of each evaluation attribute are accumulated.

The upper limit of 60% of the highest score was taken as the threshold value, i.e., 5. If the total score was lower than 5, the corresponding LF was eliminated.

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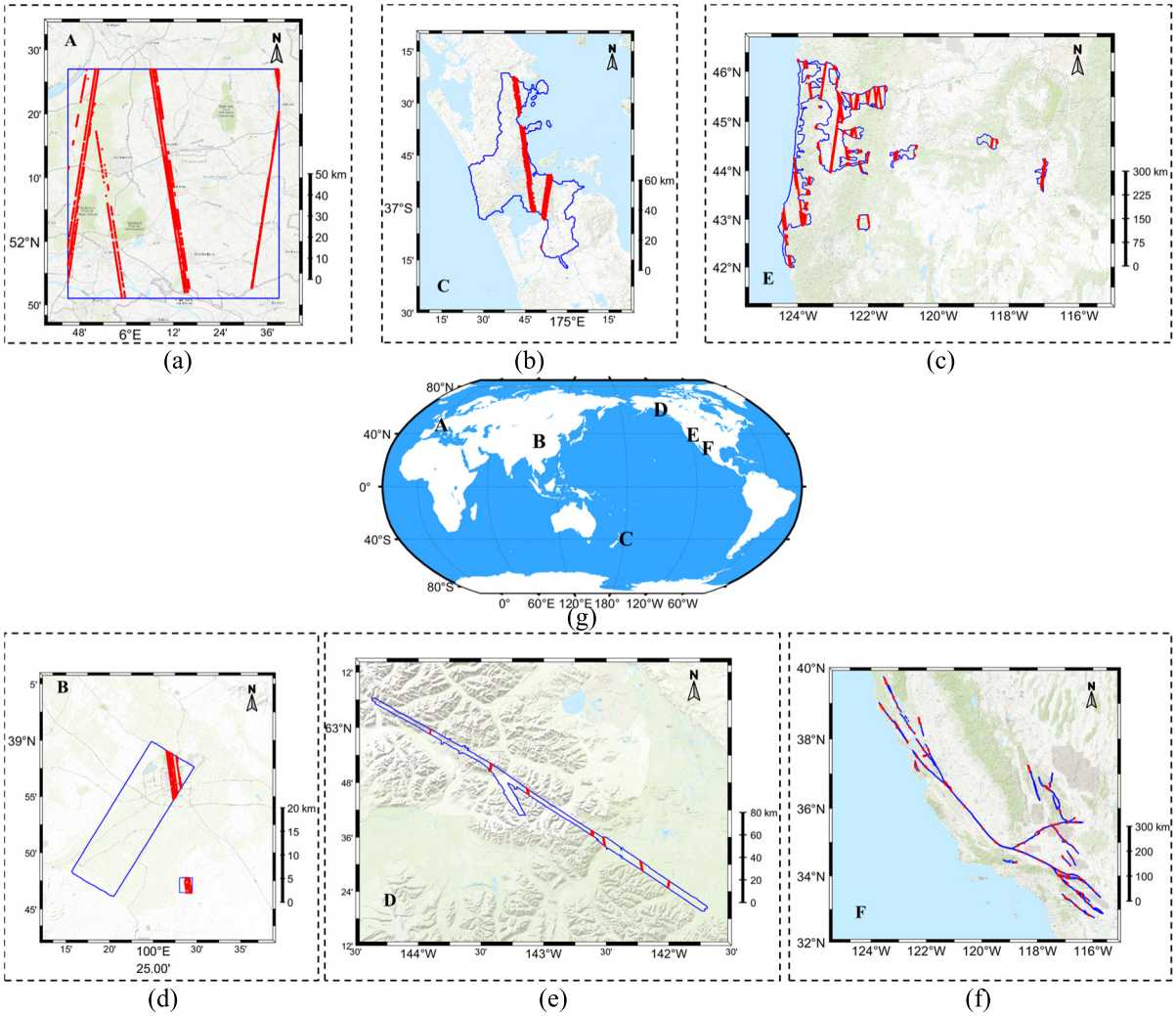


Fig. 3. Spatial distribution of validation LiDAR datasets. (a)–(f) Study areas. (g) Geolocation of the study areas. The areas surrounded by the blue lines represent the airborne LiDAR coverage areas, and the red points indicate the satellite laser altimetry data.

B. Validation of Elevation Accuracy

For the verification of LF elevation, the average of the LiDAR elevation distributed in the LF is compared with the footprint elevation of which the echo had one peak, and the average of the LiDAR ground elevation distributed in the LF is compared with the LF elevation of which the echo had multiple peaks [16]. The LiDAR ground elevation can be selected from LiDAR data of which the category label was "ground."

In the Zhangye area of China, i.e., *B* area of Fig. 3, due to urban development, some new facilities have been built in the urban areas after the satellite laser altimetry data survey time. Meanwhile, there are also obvious changes in vegetation height in some areas. To avoid any impact on the verification results, the verification points in the areas with new facilities were removed, while for the verification points in areas with obvious vegetation height changes, only the LiDAR data belonging to the "ground" category label were retained as the comparison data. These two types of data represent about 3.8% and 3.5%.

The coordinate systems of three data sources, i.e., ICESat altimetry data, DSM data, and airborne LiDAR data, are inconsistent. In data processing, the data coordinate systems need to unify. To improve the data processing efficiency, the LF elevations were converted to the airborne LiDAR data coordinate system for the validation of LF elevation and were converted to the DSM data coordinate system when compared with the DSM data. The conversations of the height datum and planimetric coordinates were based on [35]–[40].

III. Experimental Design and Data

A. Experimental Design

The experiments of analyzing and validating the method are mainly designed in three parts. The first part is to analyze the global elevation control point dataset constructed by this method, including its distribution and number statistics, etc. The second part is to use the airborne LiDAR data collected from six validation regions to validate the accuracy of LF elevations extracted by the proposed method. Fig. 3 shows these regions' distribution. In this part, two extracted results of other methods are used to compare with that of the proposed method. The third part is to analyze the results extracted from LFs with the correction of the planimetric geographic deviation.

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TABLE I
PARAMETERS OF AIRBORNE LIDAR DATA

Aera	Acquisition time period	Point density pts/m2	Datum	Vertical RMSE (cm)
A	2007–2012	6–10	Amersfoort/RD New	10
B	2012	~4	UTM/WGS84 ellipsoid	~7-16
C	2013	4.17	NZTM2000, NZGD2000/NZVD2009	10
D	2008	5.68	UTM/WGS84 ellipsoid	~5
E	2007–2010	10.30	Oregon Lambert NAVD83/NAVD88	~7
F	2005–2007	2.98–5.17	UTM/WGS84 ellipsoid	~5-10 (some hilly or mountainous areas: 5–30)

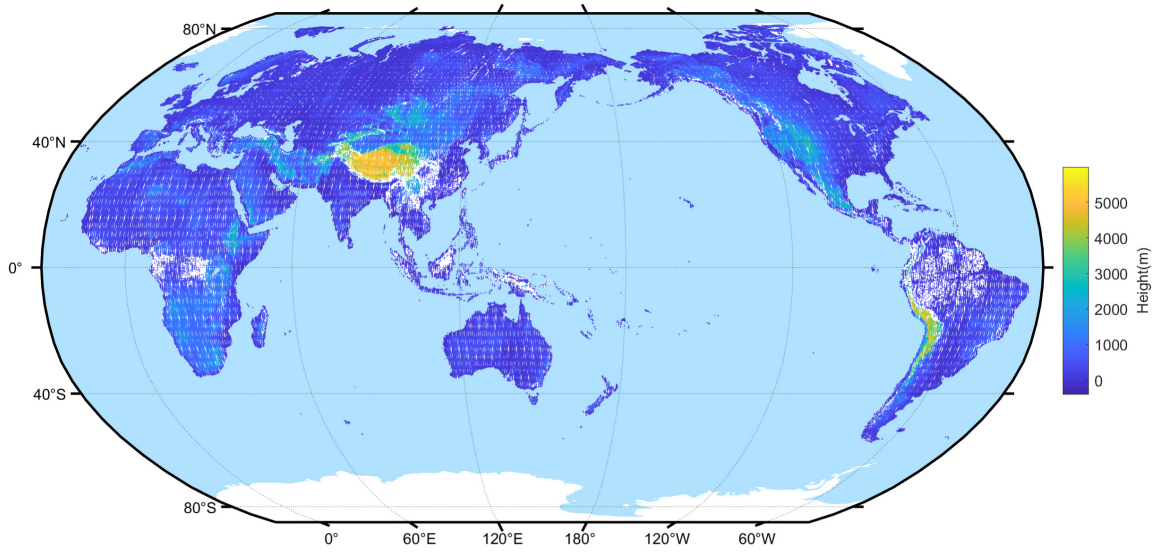


Fig. 4. Global distribution of the high-accuracy LF extraction results.

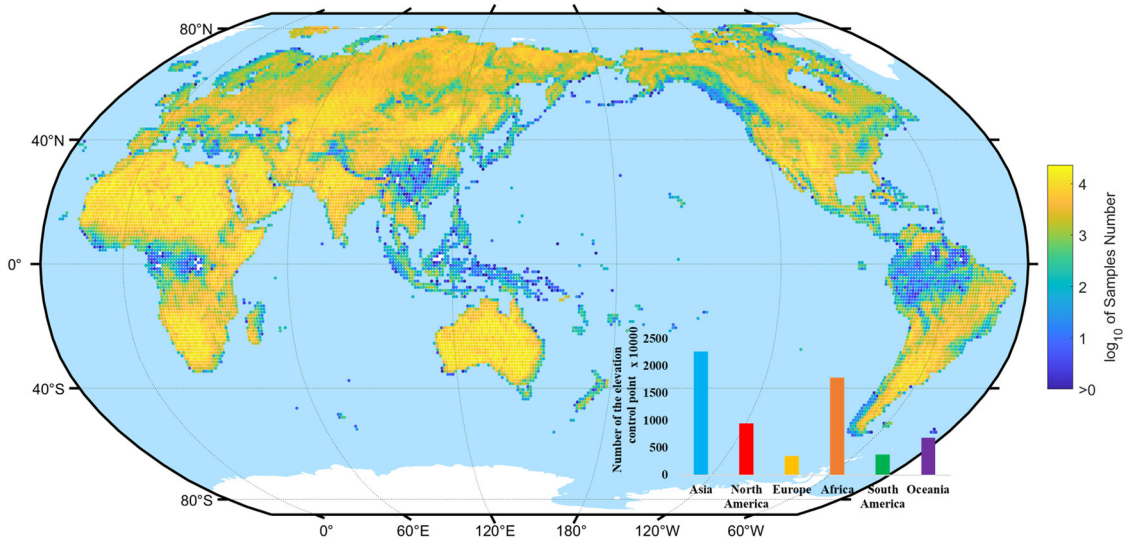


Fig. 5. Log-scaled number of high-accuracy LF extraction results within each 1° × 1° area.

B. Experimental Data

1) Satellite Laser Altimetry Data: The ICESat GLAH01 and GLAH14 products from 2003 to 2009, spatial coverage from 86 °N to 86 °S, were used for the extraction of the global LF elevation control points. The GLAH01 and GLAH14 products mainly include raw laser waveform data and surface elevations, respectively. The datum of the surface elevations is the TOPEX/Poseidon ellipsoid.

2) Digital Surface Model (DSM): The World 30 m (AW3D30) DSM is used as the data source for the calculation of terrain attributes such as slope and roughness. DSM is also used for removing the LF outliers, as

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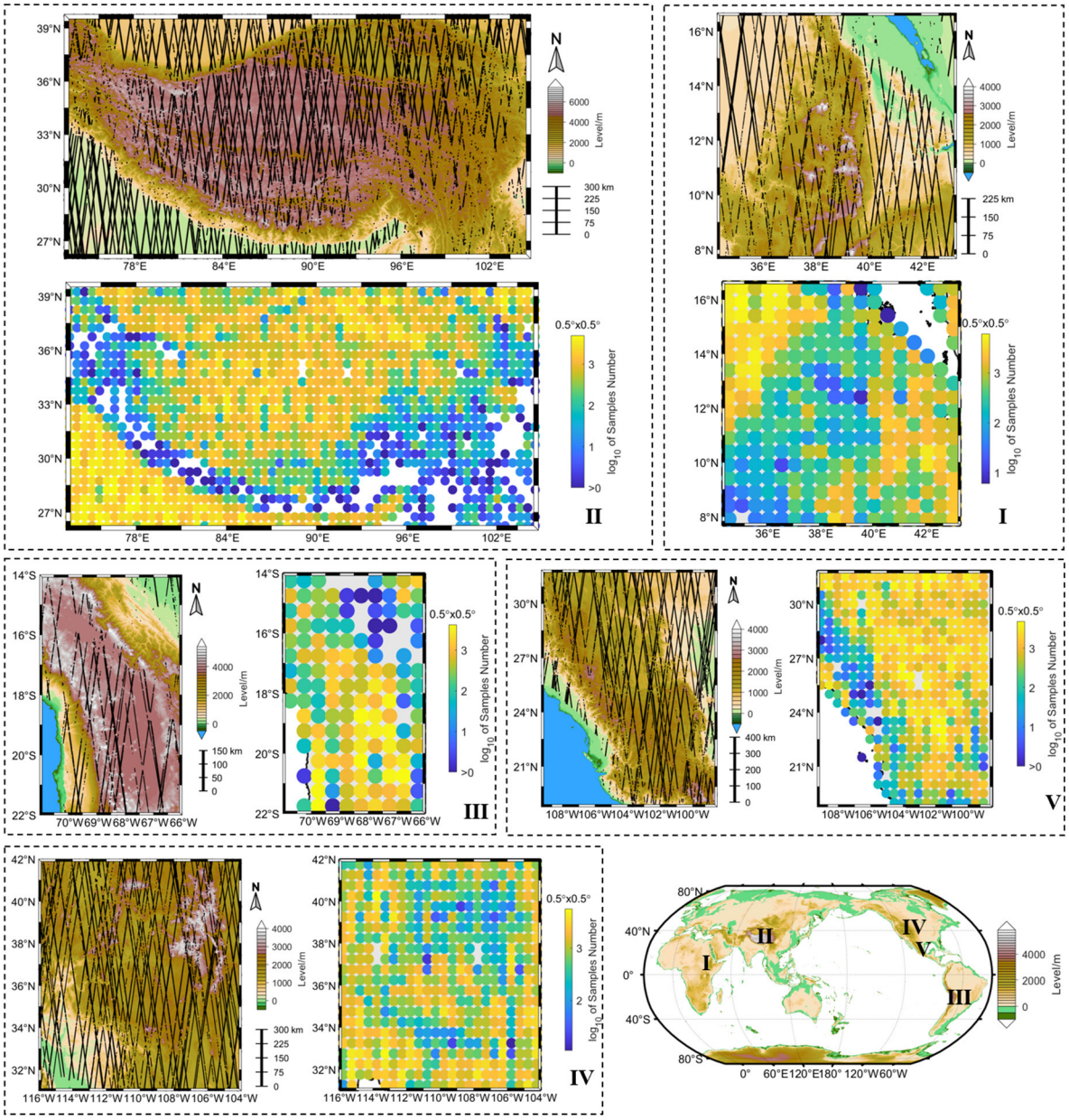


Fig. 6. Distribution of the high-accuracy LF extraction results in some plateau and mountainous areas.

presented in Section II-B. The root-mean-square error (RMSE) of AW3D30 is ∼5 m, and its geoid height is the geodetic reference system 1980 ellipsoid model with the ITRF97 [21].

3) Airborne LiDAR Data: The airborne LiDAR datasets with the centimeter/submeter vertical accuracy were used for validation. Six LiDAR datasets around the world were used (see Fig. 3). The data were selected based on the following criteria: 1) the point cloud has been classified into different categories, such as "ground"; 2) the data are georeferenced and have an accuracy report; 3) the acquisition time gap between LiDAR and laser altimetry is less than five years; and 4) the area of the data is larger than >200 km^2.

According to the above criteria, airborne LiDAR data from six areas were selected, i.e., Area A [22], Area B [23], [24], Area C [25], Area D [26], Area E [27], and Area F [28]–[30]. The data from China, i.e., Area B in Fig. 3, can be downloaded from the National Tibetan Plateau Data Center. Other airborne LiDAR data can be obtained from some free data download services, such as Open Topography, the United States Geological Survey National Map Services, the National Oceanic and Atmospheric Administration Digital Coast service, and the *Actueel Hoogtebestand Nederland* (AHN) project. The parameters of the above airborne LiDAR data are shown in Table I.

According to standards [4], [5], they can be used for validation as their vertical accuracies are at least three times better than the required accuracy.

4) Landcover Dataset: The 10-m spatial resolution landcover dataset (FROM-GLC10) was used to eliminate the elevation control points located in the water and

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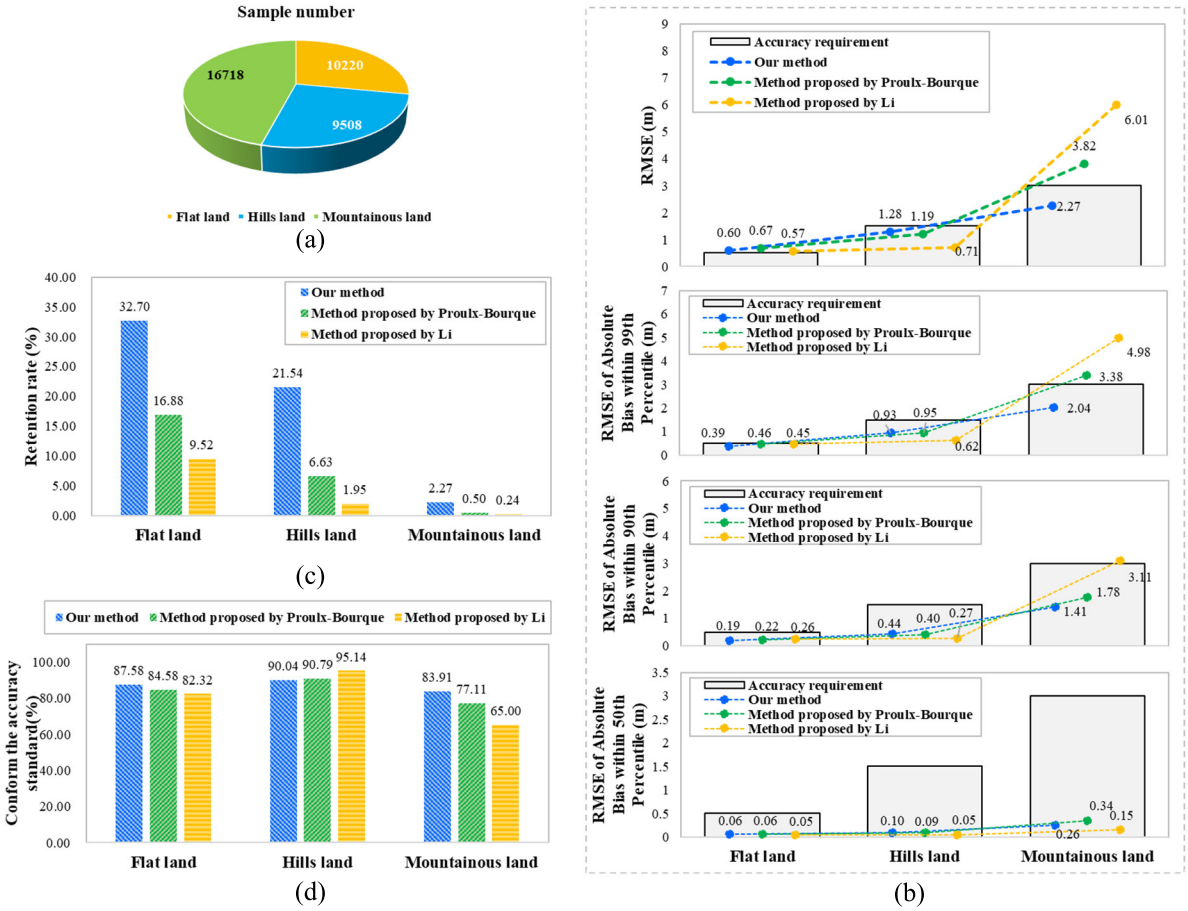


Fig. 7. Analysis of the three methods for extracting high-accuracy LFs, where (a) represents the sample numbers of the original LFs in the three different categories of terrain, and (b)–(d), respectively, represent the RMSE, the retention rate, and the proportion of extracted results that conform to the accuracy standard of high-accuracy LF elevation.

snow/ice areas. The classification accuracies of water and snow/ice have been reported to be ∼92.32% and ∼71.17%, respectively [31].

IV. Results and Discussion

Based on the definition of terrain [41], the terrain where an LF is located can be divided into the three different categories of terrain: flat (slope is less than 2°), hilly (slope is 2°–6°), and mountainous (slope is 6°–25°).

A. Distribution of the High-Accuracy Global Elevation Control Point Dataset

According to the method presented in Section II, the high-accuracy global elevation control point dataset (HAGECPD) could be extracted from the ICESat GLAH14 product. Based on FROM-GLC10, the elevation control points located in the water and snow/ice areas were eliminated. Moreover, in Greenland and Antarctica, the elevation control points located in the ice sheet area were eliminated because of the uncertainty in elevation caused by ice velocity, snow, and so on. To maximize the use of data, all the satellite laser altimetry data were processed, to generate a high-accuracy global elevation control point dataset, and the time tags that corresponded to the extracted results were reserved for users to select. The verified samples accounted for about 0.056% of this whole dataset, and their validation results are shown in the next section.

After the above processing, the distribution of the global elevation control point dataset is shown in Fig. 4. In the dataset, the proportions of elevation control points located in flat, hilly, and mountainous areas were 73.94%, 23.68%, and 2.38%, respectively. According to the ASTER Global Digital Elevation Model [43], the area with 6°–25° slope accounts for approximately ∼93.81% of the world. The number (log-scaled) of high-accuracy LF extraction results within each 1° × 1° area is shown in Fig. 5. For the number of extracted results within each continent, the number of high-accuracy LF extraction results in Asia is the largest, followed by Africa, North America, Oceania, South America, and Europe, which is mainly dependent on the area, terrain, landcover, and so on of each continent.

The distribution of the dataset in some plateau areas with mountains or canyons was further analyzed. These selected areas were the Abyssinian Plateau, the Qinghai–Tibet Plateau, the Bolivian Altiplano, the Colorado Plateau, and the Mexican Plateau, as shown in Areas I, II, III, IV, and V of Fig. 6, respectively. In each area, there are two panels, where one

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TABLE II
DATA FORMAT OF THE HIGH-ACCURACY GLOBAL ELEVATION CONTROL POINT DATASET (HAGECPD)

No.	Label	Datatype	Description
1	Time	Double	Each shot's transmit time that counted in second since January 1, 2000 12: 0: 0 (UTC) with leap seconds correction.
2	ID	Uint32	A record index
3	Shot	Uint8	Identifies each laser shot within a record index.
4	Lon	Double	The geodetic latitude of the LF (TOPEX/Poseidon ellipsoid)
5	Lat	Double	The geodetic longitude of the LF (TOPEX/Poseidon ellipsoid)
6	Elevation	Double	Surface elevation of LF (TOPEX/Poseidon ellipsoid)
7	GElevation	Double	Ground elevation of LF (TOPEX/Poseidon ellipsoid)
8	PeakNumber	Uint8	The number of echo crests generated by Gaussian filtering
9	TopographyType	Uint8	The topography type of LF. 1, 2 and 3 represent flat, hilly and mountainous areas, respectively, corresponding to 0.5 m, 1.5 m and 3 m elevation accuracy requirements (RMSE).

TABLE III
STATISTICS OF THE EXTRACTED RESULTS IN ALL THE ACCURACY VERIFICATION AREAS

Terrain	RMSE (m)	Retention rate (%)	Proportion of extracted results that conform to the accuracy standard (%)
Flat	0.60	32.70	87.58
Hilly	1.28	21.54	90.04
Mountainous	2.27	2.27	83.91

represents the distribution of the LF extraction results and the other represents the distribution of the number (log-scaled) of LF extraction results within each 0.5° × 0.5° area. In these areas, the laser altimetry footprints are sparse in some of the mountain range areas, but some laser altimetry footprints can still be retained.

Each high-accuracy global elevation control point includes nine labels. The parameters of each label, i.e., the dictionary of the high-accuracy global elevation control point dataset, are listed in Table II.

The HAGECPD dataset is available for download as a single .mat format file via the National Tibetan Plateau Data Center (http://data.tpdc.ac.cn/en/, last access: September 28, 2021): https://doi.org/10.11888/Geogra.tpdc.271727 [42].

B. Accuracy Verification of the Extracted High-Accuracy Laser Footprint Elevation Results

The ICESat altimetry data distributed in the six areas shown in Fig. 3 were selected as verified samples, for which the survey times were from 2004 to 2008. The working of the ICESat altimeter in this period was relatively stable. These validation samples can be obtained by filtering the geographic coordinates and time label in the GLAH14 product.

According to the 1:10 000 topographic mapping accuracy requirements, the verified samples were processed by adopting the method elaborated in Section II to extract the high-accuracy LF elevation. Tables III and IV list the statistics of the extracted LF elevations. Table V provides an analysis of the accuracy of the original LF elevation values. Compared with the original LF elevation values, the accuracy of the LF elevation extracted by the proposed method is significantly improved. The proportion of the footprint elevation values conforming to the accuracy standards is increased from ∼26%–68% to ∼83%–90%. The RMSE of the LF elevation values is also decreased from several hundred meters to less than 3 m.

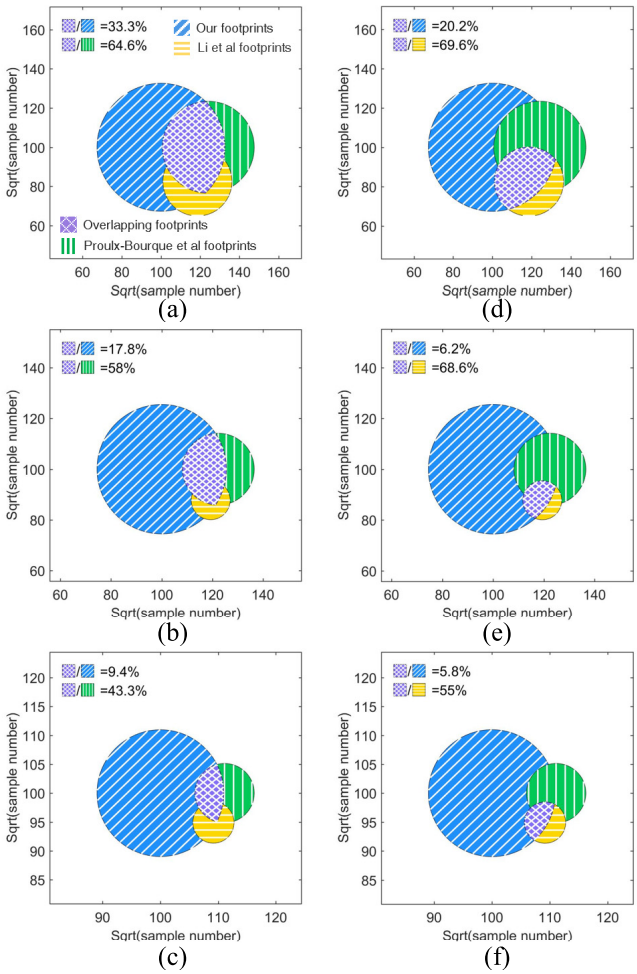


Fig. 8. Inclusion relations of the extracted footprints of the three different methods, where (a) and (d), (b) and (e), and (c) and (f) indicate the analyses about the inclusion relations of the extracted footprints of the three different methods in flat, hilly, and mountainous areas, respectively. The circle area indicates the number of samples extracted by different methods.

To further analyze the performance of the method proposed in this article, we added two other methods for comparison. These methods were proposed by Proulx-Bourque *et al.* [18] and Li *et al.* [19]. The analysis of the three methods for extracting high-accuracy LF elevation in all the accuracy verification areas is shown in Fig. 7. Compared with the original LF elevation, the accuracy of the LF elevation extracted by

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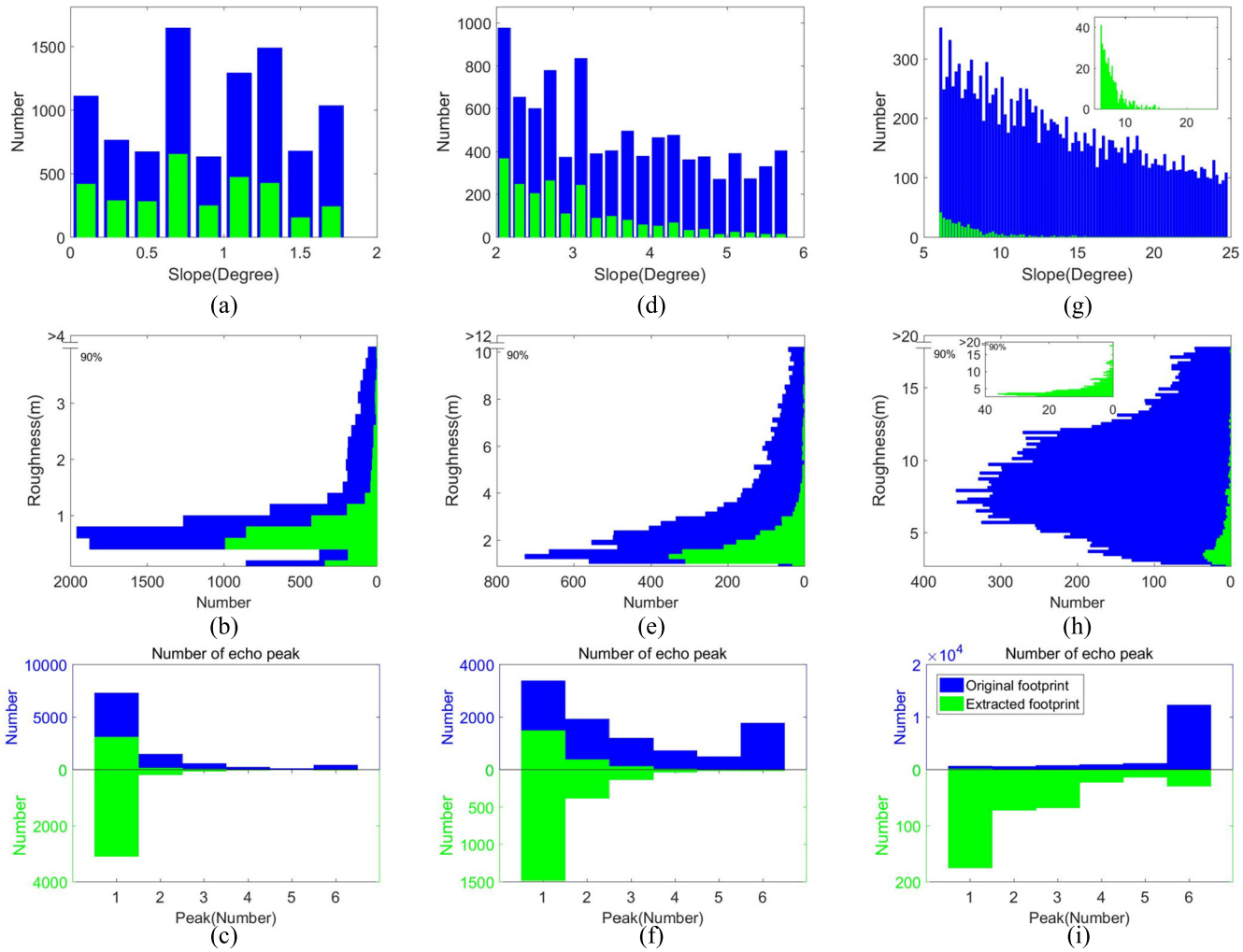


Fig. 9. Slope, roughness, and peak statistics of the original and extracted footprints, where (a)–(c), (d)–(f), and (g)–(j) indicate the various indicators in the flat, hilly, and mountainous areas, respectively.

TABLE IV
STATISTICS OF THE REJECTION RATIOS

Terrain	Rejection ratio (%) for the outliers	Rejection ratio (%) for the LF purification
Flat	2.24	65.79
Hilly	4.28	75.10
Mountainous	6.55	91.33

TABLE V
STATISTICS OF THE ORIGINAL LF ELEVATION IN ALL THE ACCURACY VERIFICATION AREAS

Terrain	RMSE (m)	Proportion of extracted results that conform to the accuracy standard (%)
Flat	227.96	68.24
Hilly	309.36	59.97
Mountainous	334.82	26.52

the three methods is significantly improved. The RMSE of the LF elevation is decreased from several hundred meters to less than 10 m. For the three different categories of terrain, the proportion of extracted results that conform to the accuracy standard is improved from 68.24%, 59.97%, and 26.52% to 87.58%, 90.04%, and 83.91%, respectively. Moreover, the RMSE of the results extracted by the three methods in the flat area is 0.57–0.67 m (slightly higher than 0.5 m), and the RMSE of the results extracted by the three methods in the hilly area is 0.71–1.28 m (less than 1.5 m). In the mountainous area, the RMSE of the results extracted by the proposed method is less than 3 m, while the RMSEs of the other two methods are 3.82 and 6.01 m, as shown in Fig. 7(b).

The RMSEs of the results extracted by the three methods in the flat area do not strictly meet the standard of being less than 0.5 m. We found that the inconsistency in the data's survey time range between the airborne LiDAR and ICESat may have caused this phenomenon. The RMSEs of the results extracted by the three methods in the flat areas of Areas A–F are 0.66–0.91, 0.67–0.7, 0.87–0.93, 0.54, 0.35–0.64, and 0.17–0.4 m, respectively, corresponding to survey times for the airborne LiDAR data of 2007–2012, 2012, 2013, 2008, 2007–2010, and 2005–2007, respectively. Moreover, since the accuracy requirement for the footprint elevation in the flat area is higher than that in the other two kinds of terrain, the

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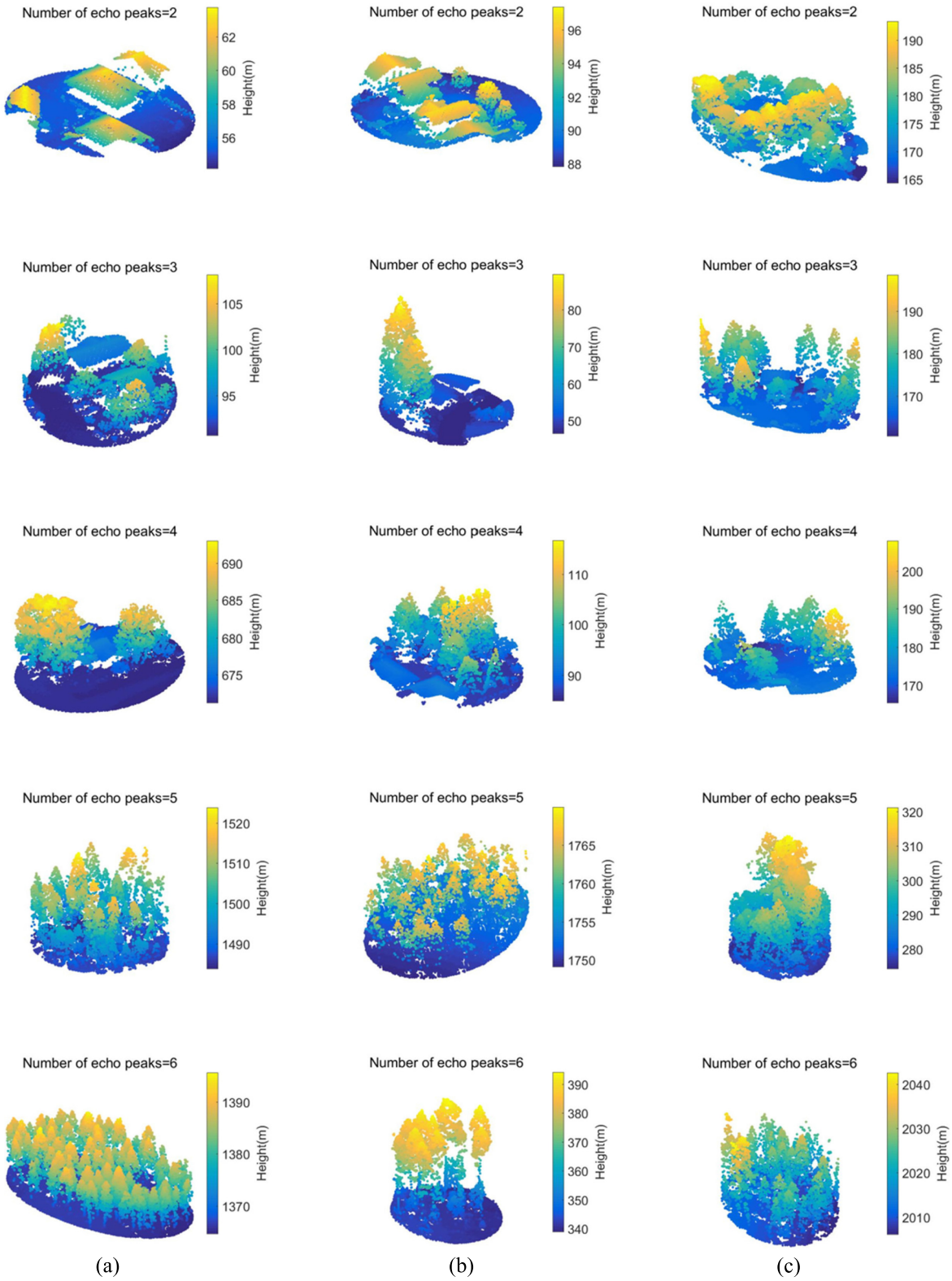


Fig. 10. Instances of extracted footprints of which echo has multiple peaks, where (a), (b), and (c), respectively, indicate a stereograph of the LiDAR data within the LF in flat, hilly, and mountainous areas.

different acquisition times between the two data sources have a greater impact on the correctness of the verification results of footprint elevation in the flat area. To mitigate the impact of this factor, RMSEs of absolute bias with 99th, 90th, and 50th proportion of extraction results were analyzed, as shown in Fig. 7(b). The RMSE of absolute bias with the 99th proportion of three extraction results in the flat area conforms to the accuracy standards. Overall, the LF elevation extracted by the proposed method can conform to the accuracy standards in all three kinds of terrain, while the LF elevation extracted by the

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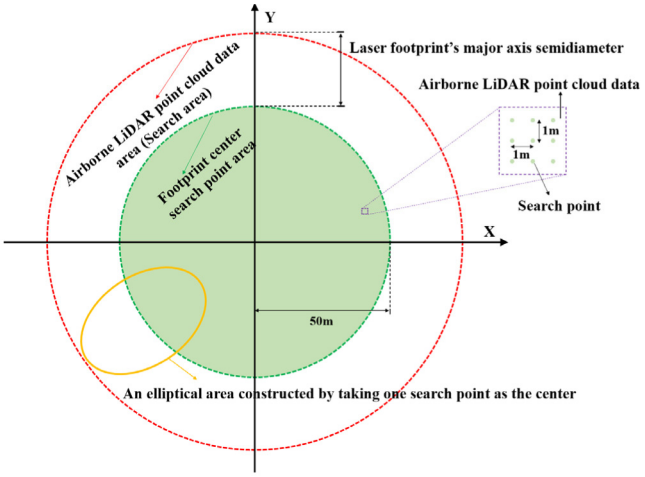


Fig. 11. Sketch of the LF search area.

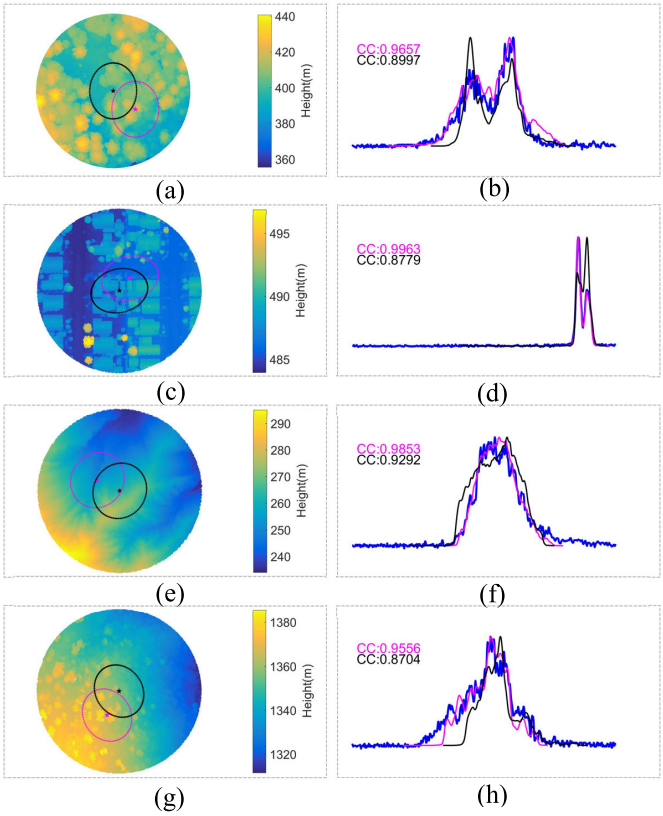


Fig. 12. Example of improving the footprint planimetric geolocation by the waveform matching method. (a), (c), (e), and (g) Search areas. The area enclosed by the black line and the area enclosed by the purple line represent the GLAH14 LF and the best-match LF, respectively. The black pentagram and the purple pentagram represent the planimetric geolocation of the GLAH14 LF and the best-match LF, respectively. (b), (d), (f), and (h) Waveform matching results. The blue line, black line, and purple line represent the GLAH01 echo waveform, the GLAH14 footprint simulated echo waveform, and the best-match footprint's simulated echo waveform, respectively. CC represents the correlation coefficient of two types of echoes (simulation and GLAH01).

other two methods can conform to the accuracy standards in the flat and hilly areas.

For the retention rate, in the three different categories of terrain, our result is the highest, and the result of Li *et al.* is the lowest, as shown in Fig. 7(c). For the three different categories of terrain, the retention rates of the footprint elevation extracted by the proposed method are 32.7%, 21.54%, and 2.27%, while those obtained by the other two methods are 16.88%, 6.63%, and 0.50%, and 9.52%, 1.95%, and 0.24%. For the proportion of extracted results that conform to the accuracy standard, our result is the highest for the LT elevations that belong to the flat and mountainous categories, and the results of Li *et al.* are the highest for LT elevations that belong to the hilly categories, as shown in Fig. 7(d). Based on the above analysis, among the three methods, our method can assure that the proportion of extracted results that conform to accuracy standards is either very close to or better than that of the other two methods, with a much better performance of footprint retention rate.

For the inclusion rates of the extracted footprints of the three different methods, the results of the analysis are shown in Fig. 8. In the three different kinds of terrain, between the proposed method and the method purposed by Proulx-Bourque *et al.*, the proportion of overlapping footprints relative to the footprints extracted by Proulx-Bourque *et al.* is higher in the flat and hilly areas (58%–64.6%) and is lower in the mountainous area (43.3%).

A similar phenomenon also appears in the analysis of the inclusion relations between the proposed method and the method of Li *et al.* This is primarily because the mountainous area footprints extracted by the methods of Proulx-Bourque *et al.* and Li *et al.* have lower proportions of extracted results that conform to the accuracy standards (77.11% and 65%), while those extracted by the proposed method have a higher proportion of extracted results that conform to the accuracy standards (83.91%), as shown in Fig. 7. To further analyze our method, histograms of the original footprints and the footprints extracted by the proposed method with respect to terrain or surface attributes, i.e., slope, roughness, and echo peaks' number were generated, as shown in Fig. 9. The extracted footprints are mainly distributed in the flatter and smoother Earth's surface (slope $<15°$ and roughness <10 m). For the number of echo waveforms, most of the extracted footprints have only a single peak, but some of them are still multipeak or even six-peak footprints, which means that the proposed method can extract footprints that conform to the accuracy standards in more complex surface areas, such as urban or forest areas. Some multipeak footprint examples that conform to the accuracy standards in flat, hilly, and mountainous areas are shown in Fig. 10. In the topographic mapping, the geographic location of the LF center can be used to locate the LF elevation in the stereoscopic image, and the area where the LF elevations are distributed can be analyzed based on the information of the stereo image. The LF elevations can be directly used as the elevation control point within the areas (bare, desert, ice sheet, and so on) where the ground objects in the vertical direction are uniformly distributed. The echo of LF in the area is one peak. The ground elevation of LFs can be used as the elevation control point within the area (urban, forest, and so on) where the ground

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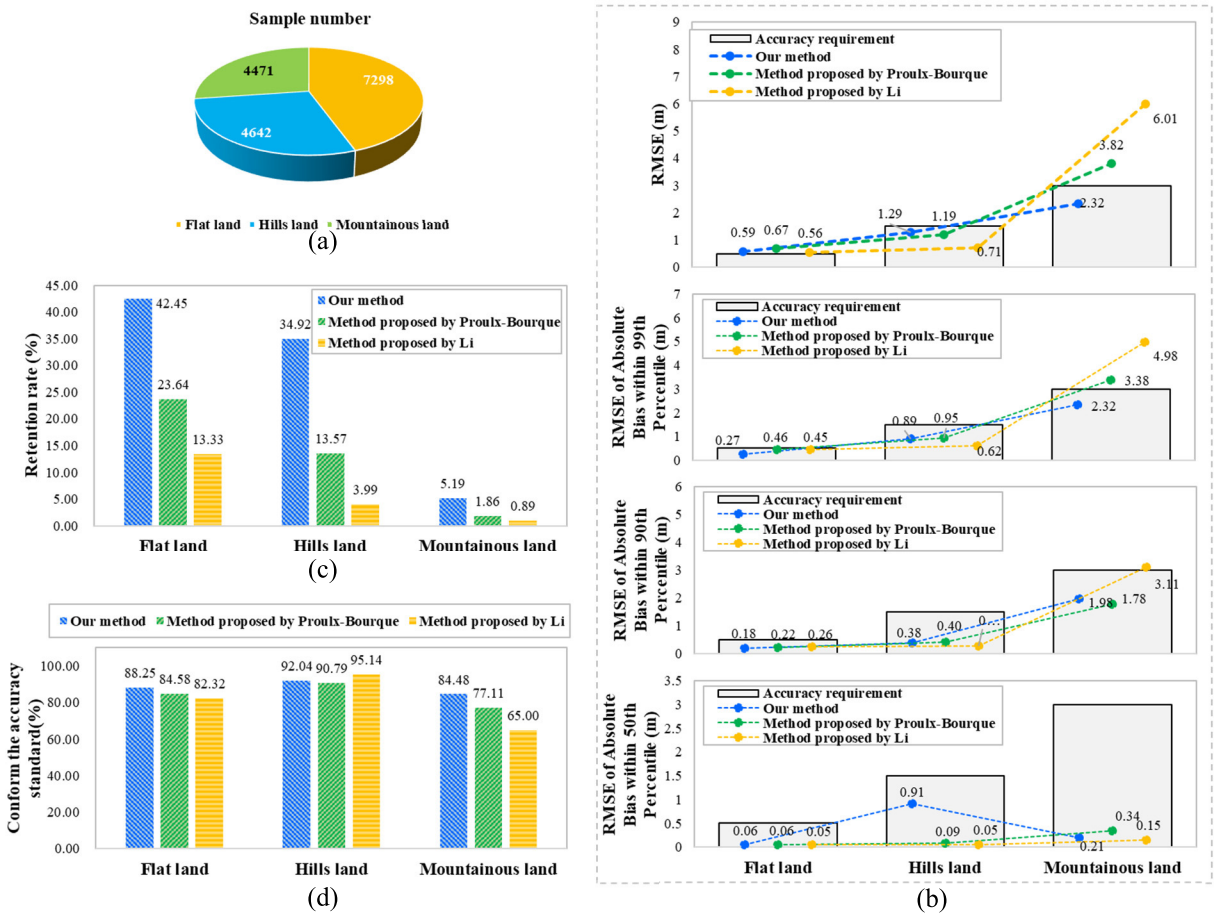


Fig. 13. Analysis of the three methods for the extraction of high-accuracy LFs from LFs that had been processed by planimetric correction in the three different categories of terrains, where (a) represents the sample number of the original LFs that had been processed by planimetric correction, and (b)–(d), respectively, represent the RMSE, the retention rate, and the proportion of extracted results that conform to the accuracy standard of high-accuracy LF elevation.

objects in the vertical direction are multiple distributed. The echo of LF in the area is multipeak.

As a notable feature, the two other methods both use an elimination criterion, where only the echoes of one crest are retained. We analyzed the extracted results of the three methods using the original single-peak LF elevation. Compared to Fig. 7, the sample numbers of the original single-peak LF (flat: 7298, hilly: 3383, and mountainous: 687) clearly decrease with the increase of the surface slope, which is because the surface slope can generate multiple peaks for the echo waveforms [32]. In flat, hilly, and mountainous areas, our extracted single-peak LF elevations that conform the accuracy standard are ∼88%, ∼93%, and ∼87%, respectively; the RMSEs of these results are 0.59, 1.25, and 2.12 m, respectively. The accuracy of our results extracted from the original single-peak LF is slightly improved by comparing that our results extracted from the original LF. It shows that the accuracy of the extracted results for which the echo has a single peak is slightly higher than that for which the echo has multiple peaks, for the extraction of the original LF elevation. For the retention rate of the original single-peak LF, our results, Proulx-Bourque *et al.* results, and Li *et al.* results are ∼25%–42%, ∼12%–23%, and ∼5%–13%, respectively. Compared to Fig. 7, the retention rate of the results extracted by the proposed method is clearly improved, which shows that the proportion of extracted results for which the echo has a single peak is higher than that for which the echo has multiple peaks, for the extraction of the original LF elevation. Moreover, among the three methods, our method can also assure that the proportion of extracted results that conform to the accuracy standards is either very close to or better than that of the other two methods, with a much better performance of footprint retention rate, for the extraction of the LF elevation for which the echo has a single peak.

C. Extensibility Analysis of the Different Methods

When verifying the accuracy of the extracted footprint elevation and considering the low planimetric accuracy of the ICESat footprint (which is generally about 10-m RMSE), we did not use the airborne LiDAR data located at the LF center to compare for the validation of LF elevation, but instead used the airborne LiDAR data located in the entire footprint to compare with the LF elevation, which can reduce the difference in the verification results caused by footprint planimetric positioning error. To further reduce the effect of the footprint planimetric geolocation for the verification of the

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footprint elevation accuracy, we adopted a waveform matching method [44]–[46] to improve the footprint planimetric geolocation. The main process of this method is given as follows.

1) For each LF provided by ICESat, a circular region with a radius of 50 m, plus the LF's major axis semi-diameter, is constructed based on the planimetric geolocation of the LF, as shown in Fig. 11.
2) In the circular region, a concentric circle with a radius of 50 m is the footprint center search point area, in which the search points are evenly distributed with an interval of 1 m, as shown in the green area of Fig. 11.
3) Taking each search point as the center, an elliptical area is constructed according to the ellipse parameters (long axis, eccentricity, and direction angle) of the LF. An example of LF is shown in the orange ellipse of Fig. 11. According to the airborne LiDAR data distributed in each search point's elliptical area and the satellite laser echo simulation model [44]–[46], the corresponding simulated echo can be simulated for each search point.
4) By comparing all the search points' simulated echoes with the original echoes corresponding to the LFs, the center of the ellipse region that corresponds to the simulated echo with the highest similarity is regarded as the real planimetric geolocation of the LF. Some examples are shown in Fig. 12.

As the planimetric error of LFs in flat areas has little influence on the elevation and the fact that the waveform matching method is only applicable to areas with distinct height characteristics [46], for the LFs located in a flat area or with a single-waveform echo signal, the planimetric geographic positioning of the LF was considered to be the positioning provided in the product. Moreover, based on the related research that has shown that the ICESat planimetric deviation (RMSE) value is mainly less than 30 m [44], [46], to assure the reliability of the LF planimetric correction, we only kept the LFs for which the planimetric deviation value was less than this value after waveform matching.

After the above processing, the three methods for the extraction of high-accuracy LF elevation were used in the LFs that had been processed by planimetric correction. An analysis of the extracted results is shown in Fig. 13. From the extracted results, it is known that our proposed method can assure that the purity of extracted results is either very close to or better than the results of the other two methods, with a much better performance of footprint retention rate.

V. Conclusion

In this article, we have described how we built an altimetry dataset that combines multielevation evaluation labels and a ranging error model to acquire high-accuracy global elevation control points that conform to the accuracy standard about elevation control point of 1:10 000 topography mapping from satellite altimetry data. In six study areas around the world, we used the airborne LiDAR data as the validation data to verify the footprints extracted by the proposed method. Furthermore, the results of the methods presented by Proulx-Bourque *et al.* and Li *et al.* were added for comparison. The results of this comparative experiment indicated that the method presented in this study can assure the accuracy of extracted results with a much better performance of footprint retention rate. The proposed method can overcome the problem of previous studies that the mapping accuracy may be low in complex scenes such as urban and forest areas since its area's spatial distribution is sparse. The high-accuracy global elevation control point dataset (HAGECPD) built using the proposed method has been released openly. The dataset includes more than 60 million elevation control points in the extracted results, with the proportion of elevation control points located in flat, hilly, and mountainous areas being 73.94%, 23.68%, and 2.38%, respectively. The dataset is beneficial to the mapping of high-accuracy terrain with meter-level resolution around the world.

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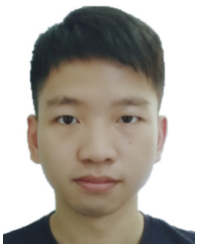
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